

Yanghao Chen

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Education

Tongji University

Bachelor of Engineering in Engineering Mechanics

GPA: 87/100

Shanghai, China

Sep. 2023 – Present; expected Jun. 2027

Johns Hopkins University

Visiting Undergraduate Research Intern, Department of Civil and Systems Engineering

Baltimore, MD, USA

Jan. 2026 – Oct. 2026

Research Interests: Solid Mechanics, Scientific Machine Learning, Neural Operators.

Research Experience

GPU-Accelerated Hybrid FEM–Neural Operator Coupling Framework

Jan. 2026 – Present

Lead Undergraduate Researcher; Advisor: [Prof. Somdatta Goswami](#)

Johns Hopkins University

- Generated displacement and stress data with GPU-accelerated JAX-FEM simulations for local neural-operator training.
- Trained Transolver models to map interface displacements to local stress fields within the neural-operator subdomain.
- Assembled interface reaction forces from predicted stresses and coupled the local neural operator with the outer FEM domain through iterative displacement–force exchange.

PINNs for Shock Capturing in the Burgers Equation

Nov. 2025 – Mar. 2026

Undergraduate Researcher; Advisor: [Prof. Xianyang \(Tom\) Chen](#)

Tongji University

- Implemented a PyTorch PINN for the one-dimensional inviscid Burgers equation, combining PDE-residual, initial-condition, and boundary-condition losses.
- Extended the baseline with constant artificial viscosity, using automatic differentiation to evaluate the derivatives in the regularized PDE residual.
- Visualized predicted shock profiles against the analytical inviscid solution, examining transition width and local deviations near the discontinuity.

ConvLSTM Modeling of Chemo-Mechanical Fields in Battery Materials

Jun. 2025 – Oct. 2025

Undergraduate Researcher; Advisor: [Prof. Ying Zhao](#)

Tongji University

- Automated polycrystalline NMC microstructure generation and MATLAB–COMSOL simulations to produce time-aligned lithium-concentration and von Mises stress images.
- Trained separate conditional ConvLSTM models to autoregressively predict RGB images of concentration or stress from past frames, grain-orientation maps, and C-rate inputs.
- Combined MSE and SSIM losses in image space with scheduled sampling and patch-based training for multi-step autoregressive prediction.

Domain-Specific LLM Fine-Tuning for Mechanics of Materials

Sep. 2024 – Apr. 2025

Undergraduate Researcher; Advisor: [Prof. Ying Zhao](#)

Tongji University

- Curated instruction–response datasets from textbooks and technical literature on stress analysis, constitutive laws, and failure theories.
- Fine-tuned Qwen2.5-7B using LoRA for question answering and concept explanation in mechanics of materials.
- Released two instruction datasets and two Qwen2.5-based model checkpoints on [Hugging Face](#).

Technical Skills

Scientific Machine Learning

PyTorch, JAX, Neural Operators (Transolver, DeepONet), PINNs, ConvLSTM

Computational Mechanics

Finite Element Methods, FEniCSx, Gmsh, COMSOL Multiphysics, ABAQUS

Scientific Computing & Tools

Python, MATLAB, Linux, Git/GitHub, LiveLink for MATLAB

LLM Fine-Tuning

Qwen2.5, LoRA/PEFT, Instruction Dataset Curation

Languages

Chinese (native), Cantonese, English